Rishi Mankala

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Education

Purdue University West Lafayette, IN
B.S. in Computer Science, concentrations in AI/ML *Expected May 2028*

- GPA: 3.72/4.0
- Leadership & Clubs: Google Development Club, ML@Purdue, Scope Consulting, Beta Chi Theta
- Relevant Coursework: Java OOP, Data Structures and Algorithms, Probability, Artificial Intelligence, Analysis of Algorithms

Technical Skills

Languages: Java, Python, Rust, C, C++, SQL, R, JavaScript/TypeScript, HTML/CSS
Libraries/Frameworks: LangChain, PyTorch, PySpark, Pinecone, FastAPI, React, Node.js, PostgreSQL, DeepEval, WebSockets
Developer Tools: AWS (Lambda, S3), Docker, Unreal Engine, Git, Linux, CUDA/MPS, GitHub Actions, Anthropic Claude
Concepts: RAG, RESTful APIs, CI/CD, Microservices, LLM Evaluation, Relational Database Design, Agile/Scrum

Experience

Air Force Research Laboratory June 2025 – August 2025
Software Engineer Intern *Dayton, OH*

- Built a RAG pipeline over damage reports using Pinecone and LangChain, reducing retrieval time by 35% for 3 analyst teams.
- Reduced repeat query latency from 2.2s to 0.6s by adding a caching layer across AWS (Lambda, S3), serving 20+ analysts daily.
- Containerized the platform with Docker, including networking, volume configs, and GitHub Actions CI/CD deployment to AFRL program leads.

London Computer Systems September 2024 – December 2024
Software Engineer Intern *Cincinnati, OH*

- Engineered a tenant portal backend in Node.js serving 10,000+ leases, lowering API response time by 32% through batching.
- Automated lease document generation for 8+ clients using AWS Lambda and S3, replacing manual templating.
- Implemented a maintenance tracking system with WebSockets and PostgreSQL, dropping ticket resolution time 40% over spreadsheet-based logging for 5 property teams.
- Migrated 10,000,000+ property records to Databricks using PySpark, consolidating storage for analytics.

Air Force Research Laboratory May 2024 – August 2024
Software Engineer Intern *Dayton, OH*

- Developed a rule-based aerial refueling simulation in C++ on Unreal Engine 5 for AFRL trainee mission training.
- Replaced manual document review with an OpenAI pipeline classifying 200+ refueling requirements, saving $10,000 for 4 AFLCMC engineers.
- Validated GPT-4 output against 40 hand-labeled refueling requirements, tuning prompts to catch misclassified edge cases.
- Packaged the requirement classifier as a CLI, allowing engineers to process new document batches without developer support.

Projects

Quantix | *Rust, io_uring, Linux Systems, SIMD* August 2026 – Present

- Designing a disk-backed vector search engine in Rust handling 10M+ vectors, aiming to achieve sub-50ms P99 latency.
- Reducing RAM overhead by developing a Scalar Quantization (SQ8) pipeline to compress HD vectors from float32 to uint8.
- Optimizing disk throughput using async I/O routines (io_uring), executing parallel vector refinement against NVMe storage.

Computer Vision Generative Models (Purdue VIP) | *PyTorch, Diffusion Models, MIDI* July 2026 – Present

- Researching a score-to-performance pipeline determining whether symbolic conditioning beats audio-only baselines for 3D hand motion, improving FID from a published 3.529 to 3.05.
- Constructing a symbolic encoder that tokenizes MIDI score events into diffusion conditioning, cutting position error 22% on a CUDA-to-MPS ported baseline.
- Cut GPU hours 50% with a staged freeze schedule while benchmarking symbolic, HuBERT, and Wav2Vec2 backends.

GuardrailAI | *Python, DeepEval, LangSmith, FastAPI, Docker, CI/CD* | GitHub July 2026 – August 2026

- Assembled an evaluation framework testing 500+ adversarial prompts across 6 failure categories using DeepEval + LangSmith.
- Identified a 23% hallucination rate on edge-case prompts, lowering violations 74% via LoRA fine-tuning on mined failures.
- Deployed the harness as a Dockerized FastAPI microservice gated in GitHub Actions with swappable local, fine-tuned, and simulated model backends.